

09 Batch reactors

Why are we learning this?

In previous lectures, we derived the batch reactor mole balance. Today we will learn how to use that balance to solve reactor design problems. We will also discover that not all batch reactors behave the same way. Depending on whether volume remains constant or changes during reaction, reactor performance can be significantly different.

By the end of today's lesson, you should understand:

- How the batch reactor mole balance is interpreted.
- How the balance is applied in reactor design calculations.
- The difference between constant-volume and constant-pressure batch reactors.
- How changing volume influences concentration and reaction rate.
- How reactor assumptions affect conversion and reaction time.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
► One Reaction: Reactor Design ◀
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
Energy Balances
↓
One Reaction: Non-Isothermal Reactor Design
↓
Multiple Reactions
↓
Reactor Safety
↓
Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

A batch reactor design problem usually asks:

How much time is required to achieve a desired conversion?

Today we learn how to answer that question.

Batch reactor mole balance:

- In terms of number of moles (N): $\frac{dN_A}{dt} = r_A V$. In words, the rate of change in the number of moles of species A changes according to the rate of generation of A times the system volume.
 - If species A is a reactant, the rate of generation of A is
 - If species A is a product, the rate of generation of A is
 - Hence the balance equation "makes sense" because it properly accounts for the rate of change in the number of moles of the species of interest.

What Does the Batch Reactor Mole Balance Mean?

The batch reactor contains no inlet stream and no outlet stream.

As a result: Accumulation = Generation

The number of moles changes solely because reaction is occurring. This makes the batch reactor one of the simplest reactor types to analyze.

Why the Signs Make Sense

If species A is a reactant:

A is consumed so $r_A < 0$ and the number of moles decreases.

If species A is a product:

A is generated so $r_A > 0$ and the number of moles increases.

The mole balance automatically captures the correct physical behavior.

- In terms of conversion (X): $\frac{dX_A}{dt} = \frac{-r_A V}{N_{A0}}$
 - Notice that this balance is written with respect to the rate of consumption of A, because

Why Use Conversion Instead of Moles?

Engineers are usually interested in reactor performance. Rather than asking:

How many moles remain?

we typically ask:

What conversion has been achieved?

Writing the balance in terms of conversion makes reactor design calculations much more intuitive.

- Applying the batch reactor mole balances: If you are solving a reactor design problem for a prototypical batch reactor:
 - In other words, we've already derived the mole balance for the prototypical batch reactor. You do not need to rederive it on your problem solutions, except for the following circumstances:

- The problem statement asks you to derive the balance. In this case, we are wanting to test your ability to derive a balance.
- The reactor is not a prototypical batch reactor. E.g., it could be a semibatch reactor. In that case, you should derive the appropriate balances for the relevant species, making sure to state your assumptions.

Don't Re-Derive What Has Already Been Derived

For a prototypical batch reactor, the governing design equation is already known. Unless instructed otherwise, use the established batch reactor balance directly.

Re-derivation is only necessary when:

- The problem explicitly requests it.
- The reactor is not a standard batch reactor.
- Additional assumptions must be introduced.

- There are (at least) two types of batch reactors:

Two Important Types of Batch Reactors

Constant-Volume Batch Reactor: Volume remains fixed throughout the reaction. Concentration changes only because reactants are consumed and products are formed.

Constant-Pressure Batch Reactor: Volume may change during reaction. When operated in gas phase, concentration changes because:

- species amounts change, or
- reactor volume changes due to changes in the total number of moles or temperature.

These reactors often behave quite differently.

- Let's do an example: The elementary gas phase reaction $A + B \rightarrow C$ is carried out isothermally in a constant volume batch reactor and in a constant pressure batch reactor. The initial mixture is equimolar in A and B, and no C is present initially in both cases. Which reactor takes longer to get to 60% conversion?
- First: Practice with the concept! Pause the video, and try answering conceptually. Why did you choose the answer you did? How confident are you in your response?

Concept Before Computation

Ask yourself:

Will concentrations decrease faster?

Will reactor volume increase?

Will the reaction slow down?

Building physical intuition is just as important as obtaining the correct mathematical solution.

Why Might a Constant-Pressure Reactor Behave Differently?

For gas-phase reactions, maintaining constant pressure often requires the reactor volume to change. If volume increases:

Concentrations decrease.

If concentrations decrease:

Reaction rates decrease.

A slower reaction rate can increase the time required to reach a desired conversion.

- How would this answer change if this were a liquid phase reaction?

Why Liquid Reactions Are Different

For most liquid-phase reactions:

Volume $\approx$ Constant

even when reaction occurs. As a result, constant-pressure and constant-volume liquid reactors often behave very similarly. This is one reason liquid-phase reactor design is often simpler than gas-phase reactor design.

- Now, let's do the numerical solution:

Summary and Key Takeaways

Today we learned how to apply the batch reactor mole balance to reactor design problems.

Key Ideas

- ✓ Batch reactors have no inlet or outlet streams.
- ✓ The batch reactor balance relates accumulation directly to reaction.
- ✓ Reactor performance is often described using conversion rather than moles.
- ✓ Standard batch reactor design equations generally do not need to be re-derived.
- ✓ Constant-volume and constant-pressure batch reactors can behave differently for gas-phase reactions.
- ✓ Volume changes alter concentration, which alters reaction rate and reactor performance.

Most Important Idea

Reactor Design = Reactor balance + Rate law + Relate concentration to conversion. The goal is usually to determine the time required to achieve a desired conversion.